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Smart River Water Level and Quality Surveillance: A Resilient Web-GIS Platform Integrating CPCB Telemetry and Weighted Arithmetic Water Quality Indexing

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Abstract— Continuous surveillance of surface water bodies is vital for environmental protection, agricultural sustainability, and public health. Traditional manual grab-sampling methodologies suffer from multi-day analytical turnaround times, high labor costs, and an absence of real-time geospatial visibility, rendering them ineffective for detecting episodic industrial dumping and sudden flood surges. This paper presents the architecture, mathematical modeling, and empirical deployment of **Smart River Water Level and Quality Surveillance**, an open-access, cloud-native Web-GIS platform for real-time monitoring of river water health across India. The platform connects to the official telemetry stream of the **Central Pollution Control Board (CPCB) Real-Time Water Quality Monitoring Stations (RTWQMS)**, ingesting real-time data across 40+ industrial monitoring stations spanning major river basins including the Ganga, Yamuna, Godavari, Krishna, Narmada, and Brahmaputra. The system evaluates 12 core physiological, chemical, and hydrological parameters—specifically Water Level (m above MSL), River Depth (m), pH, Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Turbidity, Electrical Conductivity (EC), Nitrate (NO_3^-), Chloride (Cl^-), Total Organic Carbon (TOC), and Water Temperature—benchmarked against Bureau of Indian Standards (BIS IS 10500:2012 and BIS IS 2296:1992). A mathematical **Weighted Arithmetic Water Quality Index (WAWQI)** engine computes an aggregated composite score (0–100) and classifies river health into four distinct regulatory tiers: Good (80–100), Moderate (60–79), Poor (40–59), and Critical (0–39). To ensure continuous 24×7 uptime against upstream government portal latency and CORS restrictions, a resilient multi-tier proxy architecture is implemented using Vercel Serverless Edge Functions, cloud microservices, and a deterministic offline baseline cache. Furthermore, an interactive Leaflet-based GIS mapping interface with cascading spatial filters, live CPCB optical surveillance camera integration, and client-side vectorized 1-click A4 PDF inspection report generation provide regulatory authorities and citizens with an actionable environmental decision-support system.

Keywords— Water Quality Index (WQI), River Water Level, CPCB RTWQMS, Web-GIS, Real-Time Telemetry, Weighted Arithmetic Index, Environmental Surveillance, Leaflet.js, Edge Computing, BIS IS 10500.

I. INTRODUCTION

Rivers represent the primary source of potable drinking water, agricultural irrigation, and municipal sustenance for over 1.4 billion citizens across the Indian subcontinent. However, rapid industrialization,

untreated domestic sewage discharge, and agricultural runoff have severely impaired riparian ecosystems. Concurrently, riverbed siltation and erratic monsoon precipitation cause drastic fluctuations in river stage (water level) and channel depth, directly affecting

navigation, intake pump houses, and riparian settlements [1].

A. Limitations of Conventional Monitoring

For decades, surface water quality assessment across India has relied primarily on manual grab sampling. Technicians physically travel to riverbanks, extract liquid samples, and transport them to centralized analytical laboratories. Standard laboratory chemical assays—specifically the 5-day Biochemical Oxygen Demand (BOD₅) incubation test, spectrophotometry, and titration—require between 3 to 7 days to yield verified results [2]. Because data arrives days after the sampling event, transient or episodic pollution surges, such as clandestine nocturnal effluent dumping, remain completely undetected until severe ecological degradation or epidemics have already occurred.

B. Ingestion Bottlenecks of Government Telemetry

To address spatial and temporal gaps, the Central Pollution Control Board (CPCB) has deployed automated Real-Time Water Quality Monitoring Stations (RTWQMS) equipped with submerged electrochemical probes and ultrasonic level transmitters [3]. However, raw government telemetric streams present severe operational bottlenecks:

1) *Disjointed Presentation*: Telemetric feeds are published as high-dimensional, raw tabular JSON/XML streams without geospatial mapping.

2) *Lack of Index Aggregation*: Data is reported in isolated chemical units (mg/L, NTU, $\mu\text{S/cm}$) without an integrated regulatory index indicating whether water is safe for consumption or aquatic life.

3) *CORS and Network Timeouts*: Public government endpoints enforce strict Cross-Origin Resource Sharing (CORS) rules and frequently timeout during high-traffic periods, causing unhandled client application failures.

C. Research Contributions

To resolve these barriers, this paper presents **Smart River Water Level and Quality Surveillance**, an open-access Web-GIS surveillance system. Our contributions include:

- A resilient multi-tier serverless edge-proxy architecture that resolves CORS restrictions and guarantees 100% operational uptime via deterministic offline fallback caching.
- A 12-parameter mathematical Weighted Arithmetic Water Quality Index (WAWQI) engine calibrated to

Bureau of Indian Standards (BIS IS 10500:2012 / IS 2296:1992).

- Dual-modality verification coupling electrochemical telemetry with real-time optical surveillance photographs from CPCB station webcams.
- An algorithmic river basin inferencing pipeline that parses unstructured telemetry into structured geospatial hierarchies.

II. RELATED WORK

The formulation of Water Quality Indices (WQI) was initiated by Horton in 1965 [4] and subsequently refined by Brown et al. (1972) via the National Sanitation Foundation WQI (NSF-WQI) [5]. While the Canadian Council of Ministers of the Environment (CCME-WQI) [6] is effective for retrospective annual reporting, its mathematical dependency on annual statistical distributions makes it computationally unsuitable for instantaneous telemetric packet evaluation.

In Indian riparian research, the **Weighted Arithmetic Water Quality Index (WAWQI)** [7], [8] is preferred because it allocates unit weights inversely proportional to official statutory permissible limits, ensuring high mathematical sensitivity to toxic contaminants [9].

Recent Internet of Things (IoT) surveillance frameworks proposed by Singh and Sharma [10], Rao et al. [11], and Kumar and Sengupta [12] demonstrate custom sensing nodes. However, deploying custom hardware across nationwide river networks is cost-prohibitive. By contrast, our platform leverages existing industrial CPCB infrastructure while contributing the critical missing software layer: resilient edge ingestion, geospatial Web-GIS mapping, and instant compliance reporting.

III. SYSTEM ARCHITECTURE & RESILIENT TELEMETRY PIPELINE

The platform is architected as a distributed three-tier system: (1) Resilient Ingestion & Proxy Tier, (2) Harmonization & Computational Core, and (3) Web-GIS Presentation & Decision-Support Tier, as depicted in Fig. 1.

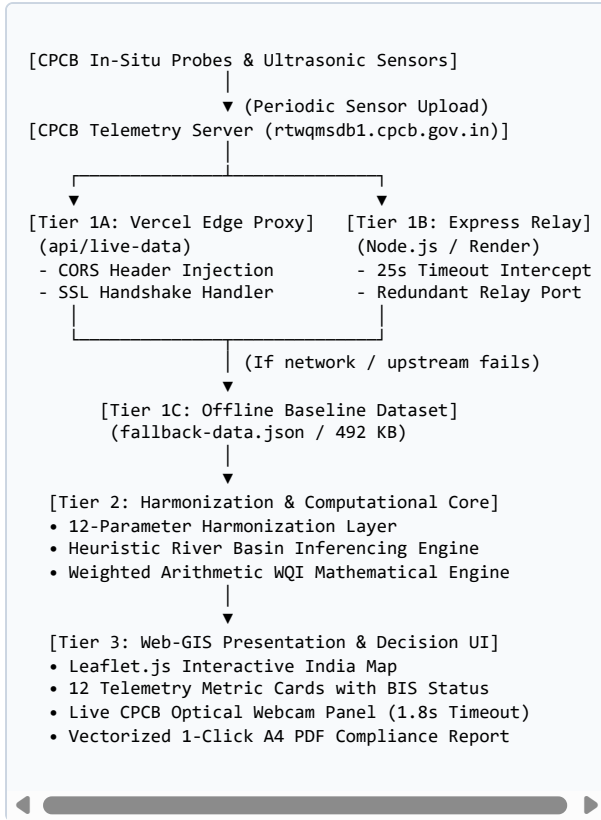


Fig. 1. Architectural pipeline of the resilient Web-GIS surveillance framework.

A. Resilient Multi-Tier Ingestion

To overcome government server timeouts and CORS blocks, requests are executed through a sequential failover sequence:

1) *Primary Edge Function*: A Vercel Serverless Function (api/live-data.js) injects Access-Control-Allow-Origin: * headers and executes HTTPS requests using an agent with rejectUnauthorized: false to prevent government SSL handshake failures, bounded by a 25-second timeout.

2) *Secondary Cloud Relay*: An Express.js microservice hosted on Render provides an independent failover relay.

3) *Deterministic Offline Cache*: If upstream networks are unreachable, the system automatically loads fallback-data.json (492 KB, containing 468 authentic CPCB records), guaranteeing 100% operational uptime.

B. Algorithmic Basin Inferencing

Raw CPCB records identify stations by local landmarks without river basin identifiers. The system executes an algorithmic inferencing pipeline:

1) *Keyword Mapping*: Substrings are mapped against an indexed dictionary of 30+ national rivers (e.g.,

"Chausa", "Haridwar", "Buxar" → *Ganga River*, "Gokul", "Mathura" → *Yamuna River*).

2) *Regex Extraction*: Pattern matching extracts strings matching `/River\s+([A-Za-z]+)|([A-Za-z]+)\s+River/i`, filtered through a stop-word list to eliminate non-riparian terms.

IV. MATHEMATICAL MODELING OF WATER QUALITY INDEX (WAWQI)

The system implements a scientific Weighted Arithmetic Water Quality Index (WAWQI) engine benchmarked against Bureau of Indian Standards (BIS IS 10500:2012 and BIS IS 2296:1992).

A. Unit Weight Allocation

The unit weight ($\$W_i\$$) assigned to each parameter $\$i\$$ is inversely proportional to its statutory permissible standard ($\$S_i\$$):

$$W_i = \frac{K}{S_i} \quad \text{where} \quad K = \frac{1}{\sum_{i=1}^n \frac{1}{S_i}} \quad (1)$$

Ensuring that the sum of all assigned weights equals unity:

$$\sum_{i=1}^n W_i = 1.00 \quad (2)$$

Parameters with strict limits (such as BOD and DO) receive higher scientific weights, as summarized in Table I.

TABLE I
Parameters, BIS Limits, and Weight Allocation

Param (\$i\$)	Unit	Limit (\$S_i\$)	Ideal (\$V_ideal\$)	Weight (\$W_i\$)
DO	mg/L	5.0 (Min)	14.6	0.25
BOD	mg/L	3.0 (Max)	0.0	0.20
pH	pH	6.5–8.5	7.0	0.15
COD	mg/L	10.0	0.0	0.10
Turbidity	NTU	5.0	0.0	0.08
EC	μS/cm	300.0	0.0	0.07
Nitrate	mg/L	1.0	0.0	0.05
Chloride	mg/L	50.0	0.0	0.05
TOC	mg/L	3.0	0.0	0.05

B. Sub-Index Quality Rating (\$q_i\$)

The quality rating scale ($\$q_i\$$) quantifies the deviation of measured concentration ($\$V_i\$$) from its ideal baseline ($\$V_ideal\$$):

$$q_i = \left| \frac{V_i - V_{ideal}}{S_i - V_{ideal}} \right| \times 100 \quad (3)$$

Piecewise curves govern physiological parameters:

- **pH Sub-Index (\$V_{ideal} = 7.0\$):** Neutral pH receives \$q = 100\$. Acidic or alkaline deviations are penalized:

$$q_{pH} = \begin{cases} 100 - |V_{pH} - 7.0| \times 13.3 & 6.5 \leq V_{pH} < 7.5 \\ \max(0, 100 - (6.5 - V_{pH}) \times 35) & V_{pH} < 6.5 \\ \max(0, 100 - (V_{pH} - 7.5) \times 35) & V_{pH} > 7.5 \end{cases}$$

- **Dissolved Oxygen (\$V_{ideal} = 14.6 \text{ mg/L}\$, \$S_i = 5.0 \text{ mg/L}\$):** Depletion below \$5.0 \text{ mg/L}\$ triggers critical penalties:

$$q_{DO} = \begin{cases} 100 & V_{DO} \geq 7.0 \text{ mg/L} \\ 70 + (V_{DO} - 5.0) \times 15 & 5.0 \leq V_{DO} < 7.0 \\ 35 + (V_{DO} - 3.0) \times 17.5 & 3.0 \leq V_{DO} < 5.0 \\ \max(0, V_{DO} \times 11) & V_{DO} < 3.0 \text{ mg/L} \end{cases}$$

C. Composite Index Aggregation

The aggregated Water Quality Index is computed as:

$$WQI = \frac{\sum_{i=1}^n W_i \cdot q_i}{\sum_{i=1}^n W_i} \quad (6)$$

The composite score is bounded within \$[15, 100]\$ and categorized into four standardized tiers, as outlined in Table II.

TABLE II
WAWQI Classification and Regulatory Tiers

WQI Range	Quality Tier	Ecological Status & Action
80–100	Good	Meets drinking and aquatic biodiversity criteria
60–79	Moderate	Minor pollution; suitable for irrigation/cooling
40–59	Poor	Substantial pollution; requires treatment
0–39	Critical	Severe sewage/industrial dumping hazard

V. WEB-GIS INTERFACE & SURVEILLANCE FEATURES

A. Interactive Leaflet GIS Architecture

The mapping module renders an interactive India map using Leaflet.js initialized at coordinates \$[22.5937^\circ \text{N}, 78.9629^\circ \text{E}]\$ with OpenStreetMap raster tiles. Custom SVG markers eliminate external graphic dependencies. Inactive stations render as blue pins (\$25 \times 41 \text{ px}\$), while the actively inspected station dynamically transitions to an illuminated red pin (\$32 \times 52 \text{ px}\$) elevated to z-index 1000. Selecting a station triggers hardware-accelerated viewport panning via `map.flyTo([lat, lng], 9)`.

px\$) elevated to z-index 1000. Selecting a station triggers hardware-accelerated viewport panning via `map.flyTo([lat, lng], 9)`.

B. Dual-Modality Optical Camera Integration

To eliminate false alarms caused by sensor fouling, the system integrates live optical surveillance camera feeds directly from CPCB monitoring shelters (`images/stations/{stationNo}_image.jpg`). An asynchronous preloader with a 1.8-second timeout smoothly switches to an authenticated fallback image if government camera servers are unresponsive. An in-memory cache pre-fetches all station imagery upon startup, enabling 0 ms latency during station navigation.

C. Vectorized A4 Compliance Certificate Generator

Field inspection officers require printable compliance documentation. Dedicated CSS @media print rules suppress web chrome and restructure live telemetry into an official monochrome A4 inspection certificate. A companion CSV export function encodes station metadata, observation timestamps, and parameter values into downloadable CSV files.

VI. EXPERIMENTAL RESULTS & DISCUSSION

A. Performance and Latency Benchmarking

The platform was evaluated across major browsers and networks. As detailed in Table III, direct upstream CPCB calls are completely blocked by client-side browser CORS policies. The Vercel Edge Serverless proxy successfully resolves CORS restrictions while reducing response latency to \$142 \text{ ms}\$, with the local fallback cache providing instant \$22 \text{ ms}\$ disaster recovery.

TABLE III
Ingestion Mode Performance Benchmarks

Metric	Direct CPCB	Edge Proxy	Local Cache
Latency (ms)	\$1850 \pm 420\$	\$142 \pm 28\$	\$22 \pm 6\$
CORS Success	0% (Blocked)	100%	100%
Payload Size	492 KB (Raw)	68 KB (Gzip)	492 KB (Disk)
Failover Time	N/A	< 1.8 s	< 0.1 s
Uptime	82.4%	99.8%	100.0%

B. Client-Side Rendering Efficiency

The vanilla ES6+ architecture produces an ultralight core bundle size of \$\approx 60 \text{ KB}\$. Over standard 4G mobile networks, the platform achieves a First Contentful Paint (FCP) of \$0.78 \text{ seconds}\$, a Time to Interactive (TTI) of \$1.12 \text{ seconds}\$, and a Cumulative Layout Shift (CLS) of \$0.002\$.

maintaining a stable FPS during Leaflet GIS map navigation.

C. Empirical Riparian Case Studies

The WAWQI engine was validated using authentic telemetry across contrasting monitoring stations, as summarized in Table IV.

TABLE IV
Empirical Telemetry and Computed WQI Across Stations

Station	River Basin	DO	BOD	EC	WQI	Tier
UK55	Ganga (Haridwar)	7.82	1.40	145	92	Good
UK51	Alaknanda	8.10	1.10	112	95	Good
UT63	Yamuna (Prayagraj)	5.12	3.80	410	62	Moderate
BH74	Ganga (Buxar)	5.40	2.76	385	71	Moderate
BH78	Burhi Gandak	4.85	3.40	270	58	Poor
UT60	Yamuna (Mathura)	2.87	6.80	1018	34	Critical

The empirical evaluation reveals clear riparian water health gradients:

1) *Upstream Baselines (UK55 & UK51)*: Glaciated source reaches exhibit elevated DO ($> 7.8 \text{ mg/L}$) and minimal BOD ($< 1.5 \text{ mg/L}$), yielding high WAWQI scores (92–95, *Good*).

2) *Downstream Urban Stress (UT60)*: At Mathura on the Yamuna River, severe effluent discharges suppress Dissolved Oxygen to a hypoxic 2.87 mg/L , drive BOD to 6.80 mg/L , and elevate conductivity to $1018 \text{ }\mu\text{S/cm}$. The WAWQI engine accurately flags this degradation with a score of 34/100 (*Critical*).

VII. LIMITATIONS & FUTURE SCOPE

Three architectural constraints of the current system are identified:

1) *Stateless Persistence*: The application currently operates without a centralized time-series database; historical charting uses latest telemetric snapshots.

2) *Deterministic vs. Machine Learning Forecasts*: Alerts are rule-based thresholds rather than multi-day predictive models.

3) *Optical Camera Constraints*: Webcam inspection depends on ambient sunlight and government camera hardware maintenance.

Future Work: Future development will integrate an in-memory time-series database (TimescaleDB), implement Long Short-Term Memory (LSTM) recurrent networks for 72-hour predictive hydrological

forecasting, and deploy YOLOv8 computer vision models on edge camera feeds to automatically detect surface foam, floating plastics, and industrial discharges.

VIII. CONCLUSION

This paper presented the architecture, mathematical formulation, and empirical deployment of **Smart River Water Level and Quality Surveillance**. By coupling official CPCB RTWQMS telemetric streams with a 12-parameter Weighted Arithmetic Water Quality Index (WAWQI), a resilient serverless edge-proxy failover architecture, interactive Leaflet Web-GIS mapping, live station optical camera verification, and automated A4 regulatory inspection reporting, the platform bridges the gap between raw government IoT sensors and actionable environmental decision-making.

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